

KIJANI KIOSK CAPSTONE

Pipeline Demonstration Evidence

Infrastructure- Track A

Demonstration Video

Loom video: <https://www.loom.com/share/07a84669875e4570b2d97c6cfd2009e6>

The Loom demonstration covers the CI/CD workflow through the production approval gate. The accompanying screenshot evidence document continues from the approval gate and provides the remaining implementation, failure/recovery, Kubernetes, Serverless, and Prometheus evidence.

Production Approval Gate

This evidence shows the production approval gate reached at the end of the successful CI/CD pipeline execution, continuing directly from the Loom demonstration.

Running staging smoke test...

[Pipeline] sh

+ set -e

+ cp /home/node/.kube/config /tmp/jenkins-kubeconfig

+ sed -i s|/home/sharon/.minikube|/home/node/.minikube|g /tmp/jenkins-kubeconfig

+ export KUBECONFIG=/tmp/jenkins-kubeconfig

+ kubectl get deployment kk-payments -n kijani-staging

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
kk-payments	3/3	3	3	8d

+ kubectl get pods -n kijani-staging -l app=kk-payments

NAME	READY	STATUS	RESTARTS	AGE
kk-payments-784bb759c7-bpq92	1/1	Terminating	12 (37m ago)	2d9h
kk-payments-784bb759c7-z5dzs	1/1	Terminating	11 (37m ago)	2d9h
kk-payments-f977d8b7-8k9cx	1/1	Running	0	33s
kk-payments-f977d8b7-cqcrj	1/1	Running	0	13s
kk-payments-f977d8b7-h7cnn	1/1	Running	0	32s

+ kubectl rollout status deployment/kk-payments -n kijani-staging --timeout=120s
deployment "kk-payments" successfully rolled out

+ echo Staging smoke test passed.

Staging smoke test passed.

[Pipeline] }

[Pipeline] // stage

[Pipeline] stage

[Pipeline] { (Approve Production Deployment)

[Pipeline] timeout

Timeout set to expire in 5 min 0 sec

[Pipeline] {

[Pipeline] script

[Pipeline] {

[Pipeline] input

Input requested

Production Approval Granted

This evidence shows the approval being granted, allowing the validated deployment process to proceed.

Staging smoke test passed. Deploy kk-payments:1.0.0-4074d70 to production

APPROVAL_REASON

Reason for approval (required for audit trail)

smoking test approved, proceed to production

Deploy

Abort

Successful Deployment

This evidence confirms that the approved deployment completed successfully in the target environment.

```
Timeout set to expire in 5 min 0 sec
[Pipeline] {
[Pipeline] script
[Pipeline] {
[Pipeline] input
Input requested
[Pipeline] echo
Production deployment approved.
[Pipeline] echo
Approved by: anonymous
[Pipeline] echo
Approval reason: staging test passed proceed to deployment
[Pipeline] }
[Pipeline] // script
[Pipeline] }
[Pipeline] // timeout
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to Production)
[Pipeline] echo
Deploying kijanikiosk-payments:1.0.0-4074d70 to production...
[Pipeline] sh
+ set -e
+ cp /home/node/.kube/config /tmp/jenkins-kubeconfig
+ sed -i s|/home/sharon/.minikube|/home/node/.minikube|g /tmp/jenkins-kubeconfig
+ export KUBECONFIG=/tmp/jenkins-kubeconfig
+ echo Creating/updating production ConfigMap...
Creating/updating production ConfigMap...
+ kubectl create configmap kk-payments-config -n default --from-literal=NODE_ENV=production --from-li
from-literal=LOG_LEVEL=warn --from-literal=APP_PORT=3001 --from-literal=MAX_CONNECTIONS=10 --dry-run=
+ kubectl apply -f -
configmap/kk-payments-config unchanged
+ echo Creating/updating production Secret...
Creating/updating production Secret...
+ kubectl create secret generic kk-payments-secrets -n default --from-literal=DB_PASSWORD= --dry-run=
+ kubectl apply -f -
```

```

deployment "kk-payments" successfully rolled out
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions) (hide)
[Pipeline] junit
Recording test results
No test report files were found. Configuration error?
None of the test reports contained any result
[Checks API] No suitable checks publisher found.
[Pipeline] cleanWs
[WS-CLEANUP] Deleting project workspace...
[WS-CLEANUP] Deferred wipeout is used...
[WS-CLEANUP] done
[Pipeline] echo
Build status changed to SUCCESS
[Pipeline] echo
Pipeline succeeded: kijanikiosk-payments version 1.0.0-4074d70
[Pipeline] echo
Docker image: sharonshaz/kk-payments:1.0.0-4074d70
[Pipeline] echo
Artifact URL: http://172.17.0.1:8081/repository/npm-kijanikiosk/
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // timeout
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
$ docker stop --time=1 e21bfa397a83664c8c4ace800781151377d058893d8dc426f9f99c4630cc9178
$ docker rm -f --volumes e21bfa397a83664c8c4ace800781151377d058893d8dc426f9f99c4630cc9178
[Pipeline] // withDockerContainer
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS

```

Deliberate Failure Introduced

This evidence shows the intentionally introduced faulty deployment condition used to demonstrate failure detection and operational recovery.

kijanikiosk-devops-foundation

Q Type to search

Pull requests 6

Agents

Actions

Projects

Wiki

Security and quality

Insights

Settings

Test/deliberate fault #5

Ready to merge

Code

Open

SharonOmbonya wants to merge 189 commits into main from test/deliberate-fault

Conversation 0

Commits 189

Checks 0

Files changed 91

+12,492

SharonOmbonya commented yesterday

Introduces a controlled Kubernetes readiness probe failure to validate the deployment monitoring and failure-handling workflow.

SharonOmbonya added 30 commits last month

Add gitignore for Terraform files

Add terraform lock file

Update .gitignore

Updated .gitignore

Complete Week 4 IaC pipeline documentation and reflection

Jenkinsfile

Jenkinsfile2

Add package.json

package.json

Add package-lock.json

Add npm scripts

Fix git availability and Jenkins variables

Reviewers

No reviews

Still in progress? Convert to draft

Assignees

No one—assign yourself

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

Notifications

Unsubscribe

You're receiving notifications because you authored the thread.

1 participant

kijanikiosk-devops-foundation

Pull requests 6 Agents Actions Projects Wiki Security and quality Insights Settings

Test/deliberate fault #5

Open SharonOmbonya wants to merge 189 commits into main from test/deliberate-fault

Conversation 0 Commits 189 Checks 0 Files changed 91

SharonOmbonya commented yesterday

Introduces a controlled Kubernetes readiness probe failure to validate the deployment monitoring and failure-handling workflow.

SharonOmbonya added 30 commits last month

- Add gitignore for Terraform files 79899c2
- Add terraform lock file 5b6fb6b
- Update .gitignore 7b4edf5
- Updated .gitignore 0d8d230
- Complete Week 4 IaC pipeline documentation and reflection ef920b6
- Jenkinsfile 3d69cbf
- Jenkinsfile2 112142e
- Add package.json d59b212
- package.json 61e1e2a
- Add package-lock.json 120a76e
- Add npm scripts d5b6cb8
- Fix git availability and Jenkins variables bf70e52

merge status

Commit message

Merge pull request #5 from SharonOmbonya/test/deliberate-fau

Extended description

Test/deliberate fault

This commit will be authored by shazanaombonya@gmail.com.

Confirm merge Cancel

track-a / #43

Console

Progress: Download Copy

```
Started by GitHub push by SharonOmbonya
Obtained Jenkinsfile from git https://github.com/SharonOmbonya/kijanikiosk-devops-foundation.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/jenkins_home/workspace/kijanikiosk-track-a
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
using credential nexus-credentials
Cloning the remote Git repository
Cloning repository https://github.com/SharonOmbonya/kijanikiosk-devops-foundation.git
> git init /var/jenkins_home/workspace/kijanikiosk-track-a # timeout=10
Fetching upstream changes from https://github.com/SharonOmbonya/kijanikiosk-devops-foundation.git
> git --version # timeout=10
> git --version # 'git version 2.47.3'
using GIT_ASKPASS to set credentials
> git fetch --tags --force --progress -- https://github.com/SharonOmbonya/kijanikiosk-devops-foundation.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url https://github.com/SharonOmbonya/kijanikiosk-devops-foundation.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 00e7ff1e2277609b50336cd180e2f02b2bde2ca9 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 00e7ff1e2277609b50336cd180e2f02b2bde2ca9 # timeout=10
Commit message: "Merge pull request #5 from SharonOmbonya/test/deliberate-fault"
> git rev-list --no-walk 4074d70fc02f8930a6f2517d4f5cb30a988bf8b8 # timeout=10
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] isUnix
[Pipeline] withEnv
onnya/kijanikiosk-devops-foundation.git
```

```
#16 unpacking to docker.io/sharonshaz/kk-payments:1.0.0-00e7ff1 0.1s done
#16 DONE 0.3s
+ echo Docker image built successfully:
Docker image built successfully:
+ docker images sharonshaz/kk-payments:1.0.0-00e7ff1
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
sharonshaz/kk-payments  1.0.0-00e7ff1     7301628ed1c7       1 second ago       181MB
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Push Docker Image)
[Pipeline] echo
Pushing sharonshaz/kk-payments:1.0.0-00e7ff1...
[Pipeline] withCredentials
Masking supported pattern matches of $DOCKER_PASS
[Pipeline] {
[Pipeline] sh
+ set -e
+ echo ****
+ docker login -u sharonshaz --password-stdin
WARNING! Your password will be stored unencrypted in /home/node/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
+ docker push sharonshaz/kk-payments:1.0.0-00e7ff1
The push refers to repository [docker.io/sharonshaz/kk-payments]
25ff2da83641: Waiting
f18232174bc9: Waiting
dd71dde834b5: Waiting
1e5a4c89cee5: Waiting
f18232174bc9: Waiting
dd71dde834b5: Waiting
1e5a4c89cee5: Waiting
25ff2da83641: Waiting
25ff2da83641: Waiting
f18232174bc9: Waiting
dd71dde834b5: Waiting
1e5a4c89cee5: Waiting
```

```
+ kubectl apply -f /tmp/kk-payments-staging.yaml
deployment.apps/kk-payments configured
+ kubectl rollout status deployment/kk-payments -n kijani-staging --timeout=120s
Waiting for deployment "kk-payments" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "kk-payments" rollout to finish: 2 out of 3 new replicas have been updated...
error: timed out waiting for the condition
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Smoke Test Staging)
Stage "Smoke Test Staging" skipped due to earlier failure(s)
[Pipeline] getContext
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Approve Production Deployment)
Stage "Approve Production Deployment" skipped due to earlier failure(s)
[Pipeline] getContext
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to Production)
Stage "Deploy to Production" skipped due to earlier failure(s)
[Pipeline] getContext
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] junit
Recording test results
No test report files were found. Configuration error?
[Checks API] No suitable checks publisher found.
[Pipeline] cleanWs
[WS-CLEANUP] Deleting project workspace...
[WS-CLEANUP] Deferred wipeout is used...
[WS-CLEANUP] done
[Pipeline] echo
Build status changed to FAILURE
[Pipeline] echo
```


Kubernetes Pod Evidence

This evidence shows the Kubernetes workload state following the faulty deployment and provides evidence of the resulting application impact.

```
Annotations: kubernetes.io/created-by: {"kind":"ReplicaSet","name":"kk-payments-57cf8b857","uid":"102440236","apiVersion":"v1"}
Status:      Running
IP:          10.244.0.236
IPs:
  IP:        10.244.0.236
Controlled By: ReplicaSet/kk-payments-57cf8b857
Containers:
  kk-payments:
    Container ID:  docker://32e37453802745a377b31761c13019fa1edca93c7ea3205ebcad09ee23bda8be
    Image:          sharonshaz/kk-payments:1.0.0-00e7ff1
    Image ID:       docker-pullable://sharonshaz/kk-payments@sha256:7301628ed1c704492d131d80cb0d1f12271f73f2c7eb5309736416f831ab4b26
    Port:          3001/TCP
    Host Port:     0/TCP
    State:         Running
      Started:     Mon, 24 Aug 2026 03:21:52 +0300
    Ready:         False
    Restart Count: 0
    Limits:
      cpu:         500m
      memory:      256Mi
    Requests:
      cpu:         100m
      memory:      128Mi
    Liveness:      http-get http://:3001/health delay=15s timeout=1s period=20s #success=1 #failure=3
    Readiness:     http-get http://:3001/badhealth delay=5s timeout=1s period=10s #success=1 #failure=3
    Environment Variables from:
      kk-payments-config  ConfigMap  Optional: false
      kk-payments-secrets Secret    Optional: false
    Environment:         <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-f5t5b (ro)
Conditions:
  Type                     Status
  PodReadyToStartContainers  True
  Initialized               True
  Ready                     False
  ContainersReady           False
  PodScheduled              True
Volumes:
  kube-api-access-f5t5b:
    Type:              Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName:      kube-root-ca.crt
```

```
sharon@sharon-HP-ProBook-440-14-inch-G9-Notebook-PC: ~/kijanikiosk-devops-foundation/week10/serverless/receipts$ kubectl get pods -n kijani-staging
NAME                                READY   STATUS    RESTARTS   AGE
kk-api-6d874fcd7d-kc7lv             1/1     Running   19 (86m ago)   8d
kk-api-6d874fcd7d-qg6mz             1/1     Running   17 (86m ago)   8d
kk-payments-57cf8b857-9c9lm         0/1     Running   0              9m20s
kk-payments-57cf8b857-m2jtt         0/1     Running   0              9m19s
kk-payments-f977d8b7-8k9cx          1/1     Running   0              27m
kk-payments-f977d8b7-h7cnn          1/1     Running   0              27m
sharon@sharon-HP-ProBook-440-14-inch-G9-Notebook-PC: ~/kijanikiosk-devops-foundation/week10/serverless/receipts$ kubectl get deployment kk-payments -n kijani-staging -o wide
NAME                                READY   UP-TO-DATE   AVAILABLE   AGE   CONTAINERS   IMAGES
kk-payments 2/3      2             2           8d      kk-payments  sharonshaz/kk-payments:1.0.0-00e7ff1
sharon@sharon-HP-ProBook-440-14-inch-G9-Notebook-PC: ~/kijanikiosk-devops-foundation/week10/serverless/receipts$ kubectl logs kk-payments-57cf8b857-9c9lm -n kijani
```

Recovery

This evidence shows the application returning to a healthy state after the faulty condition was corrected.

Fix/restore working deployment #6

Ready to mergeCode

Open SharonOmbonya wants to merge 190 commits into main from fix/restore-working-deployment

Conversation 0Commits 190Checks 0Files changed 91+12,492

SharonOmbonya commented 18 hours agoOwner

Restore the kk-payments readiness probe after the deliberate fault.

Problem

The readiness probe was changed from /health to /badhealth , causing the application to fail its health check.

Changes

- Restored the readiness probe to /health .
- No other project files were changed.

Testing

- Verified the corrected Kubernetes deployment configuration.
- Jenkins pipeline will validate the change after merging to main.
- Staging smoke test must pass before production approval.

Related

Deliberate fault demonstration and recovery for the capstone project.

Reviews

No reviews

Still in progress? Convert to draft

Assignees

No one—assign yourself

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

NotificationsCustomize

Unsubscribe

You're receiving notifications because you authored the thread.

[Actions](#) [Projects](#) [Wiki](#) [Security and quality](#) [Insights](#) [Settings](#)

working deployment #6

bony wants to merge 190 commits into `main` from `fix/restore-working-deployment`

Commits 190 Checks 0 Files changed 91

bony commented 18 hours ago

payments readiness probe after the deliberate fault.

probe was changed from `/health` to `/badhealth`, application to fail its health check.

he readiness probe to `/health`. project files were changed.

e corrected Kubernetes deployment configuration. pipeline will validate the change after merging to main. smoke test must pass before production approval.

It demonstration and recovery for the capstone project.

Commit message

Merge pull request #6 from SharonOmbonya/fix/restore-working-deployment

Extended description

Fix/restore working deployment

This commit will be authored by shazanaombonya@gmail.com.

Confirm merge Cancel

Console

Progress: [Download](#) [Copy](#)

```
Started by GitHub push by SharonOmbonya
Obtained Jenkinsfile from git https://github.com/SharonOmbonya/kijanikiosk-devops-foundation.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/jenkins_home/workspace/kijanikiosk-track-a
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
using credential nexus-credentials
Cloning the remote Git repository
Cloning repository https://github.com/SharonOmbonya/kijanikiosk-devops-foundation.git
> git init /var/jenkins_home/workspace/kijanikiosk-track-a # timeout=10
Fetching upstream changes from https://github.com/SharonOmbonya/kijanikiosk-devops-foundation.git
> git --version # timeout=10
> git --version # 'git version 2.47.3'
using GIT_ASKPASS to set credentials
> git fetch --tags --force --progress -- https://github.com/SharonOmbonya/kijanikiosk-devops-foundation.git +refs/heads/*:refs/remotes/origin/* # tim
> git config remote.origin.url https://github.com/SharonOmbonya/kijanikiosk-devops-foundation.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 72340832d893b85a88dbb9f6be10c85288855803 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 72340832d893b85a88dbb9f6be10c85288855803 # timeout=10
Commit message: "Merge pull request #6 from SharonOmbonya/fix/restore-working-deployment"
> git rev-list --no-walk 00e7ffe2277609b50336cd180e2f02b2bde2ca9 # timeout=10
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
```

```
Running staging smoke test...
[Pipeline] sh
+ set -e
+ cp /home/node/.kube/config /tmp/jenkins-kubeconfig
+ sed -i s|/home/sharon/.minikube|/home/node/.minikube|g /tmp/jenkins-kubeconfig
+ export KUBECONFIG=/tmp/jenkins-kubeconfig
+ kubectl get deployment kk-payments -n kijani-staging
NAME                READY    UP-TO-DATE    AVAILABLE    AGE
kk-payments         3/3      3             3            8d
+ kubectl get pods -n kijani-staging -l app=kk-payments
NAME                READY    STATUS      RESTARTS    AGE
kk-payments-649fc85bb-dc5r6    1/1     Running    0           12s
kk-payments-649fc85bb-mh6bs    1/1     Running    0           35s
kk-payments-649fc85bb-xjl7w    1/1     Running    0           34s
kk-payments-f977d8b7-8k9cx     1/1     Terminating    0           38m
kk-payments-f977d8b7-h7cnn     1/1     Terminating    0           38m
+ kubectl rollout status deployment/kk-payments -n kijani-staging --timeout=120s
deployment "kk-payments" successfully rolled out
+ echo Staging smoke test passed.
Staging smoke test passed.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Approve Production Deployment)
[Pipeline] timeout
Timeout set to expire in 5 min 0 sec
[Pipeline] {
[Pipeline] script
[Pipeline] {
[Pipeline] input
Input requested
```

Staging smoke test passed. Deploy kk-payments:1.0.0-7234083 to production?

APPROVAL_REASON

Reason for approval (required for audit trail)

Approved after successful staging deployment and smoke test

Deploy

Abort

tput

ta

rints

nput

erview

formation

ip

ne

ps

i

ild

```

[Pipeline] }
[Pipeline] { (Deploy to Production)
[Pipeline] echo
Deploying kijanikiosk-payments:1.0.0-7234083 to production...
[Pipeline] sh
+ set -e
+ cp /home/node/.kube/config /tmp/jenkins-kubeconfig
+ sed -i s|/home/sharon/.minikube|/home/node/.minikube|g /tmp/jenkins-kubeconfig
+ export KUBECONFIG=/tmp/jenkins-kubeconfig
+ echo Creating/updating production ConfigMap...
Creating/updating production ConfigMap...
+ kubectl create configmap kk-payments-config -n default --from-literal=NODE_ENV=production --from-literal=DB_HOST=postgres-production --fr
from-literal=LOG_LEVEL=warn --from-literal=APP_PORT=3001 --from-literal=MAX_CONNECTIONS=10 --dry-run=client -o yaml
+ kubectl apply -f -
configmap/kk-payments-config unchanged
+ echo Creating/updating production Secret...
Creating/updating production Secret...
+ kubectl create secret generic kk-payments-secrets -n default --from-literal=DB_PASSWORD= --dry-run=client -o yaml
+ kubectl apply -f -
secret/kk-payments-secrets unchanged
+ echo Preparing production Deployment...
Preparing production Deployment...
+ sed -e s|namespace: kijani-staging|namespace: default| -e s|image: .*|image: sharonshaz/kk-payments:1.0.0-7234083| k8s/kk-p
+ kubectl apply -f /tmp/kk-payments-production.yaml
deployment.apps/kk-payments configured
+ kubectl rollout status deployment/kk-payments -n default --timeout=120s
Waiting for deployment "kk-payments" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "kk-payments" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "kk-payments" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "kk-payments" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "kk-payments" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "kk-payments" rollout to finish: 2 of 3 updated replicas are available...
deployment "kk-payments" successfully rolled out
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage

```

```

deployment "kk-payments" successfully rolled out
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] junit
Recording test results
No test report files were found. Configuration error?
None of the test reports contained any result
[Checks API] No suitable checks publisher found.
[Pipeline] cleanWs
[WS-CLEANUP] Deleting project workspace...
[WS-CLEANUP] Deferred wipeout is used...
[WS-CLEANUP] done
[Pipeline] echo
Build status changed to SUCCESS
[Pipeline] echo
Pipeline succeeded: kijanikiosk-payments version 1.0.0-7234083
[Pipeline] echo
Docker image: sharonshaz/kk-payments:1.0.0-7234083
[Pipeline] echo
Artifact URL: http://172.17.0.1:8081/repository/npm-kijanikiosk/
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // timeout
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
$ docker stop --time=1 2a0eec7ad2ae51bc053868e1feb33615ce698ca86d2e58da3b4b47930113dc94
$ docker rm -f --volumes 2a0eec7ad2ae51bc053868e1feb33615ce698ca86d2e58da3b4b47930113dc94
[Pipeline] // withDockerContainer
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS

```


Kubernetes Recovery Evidence

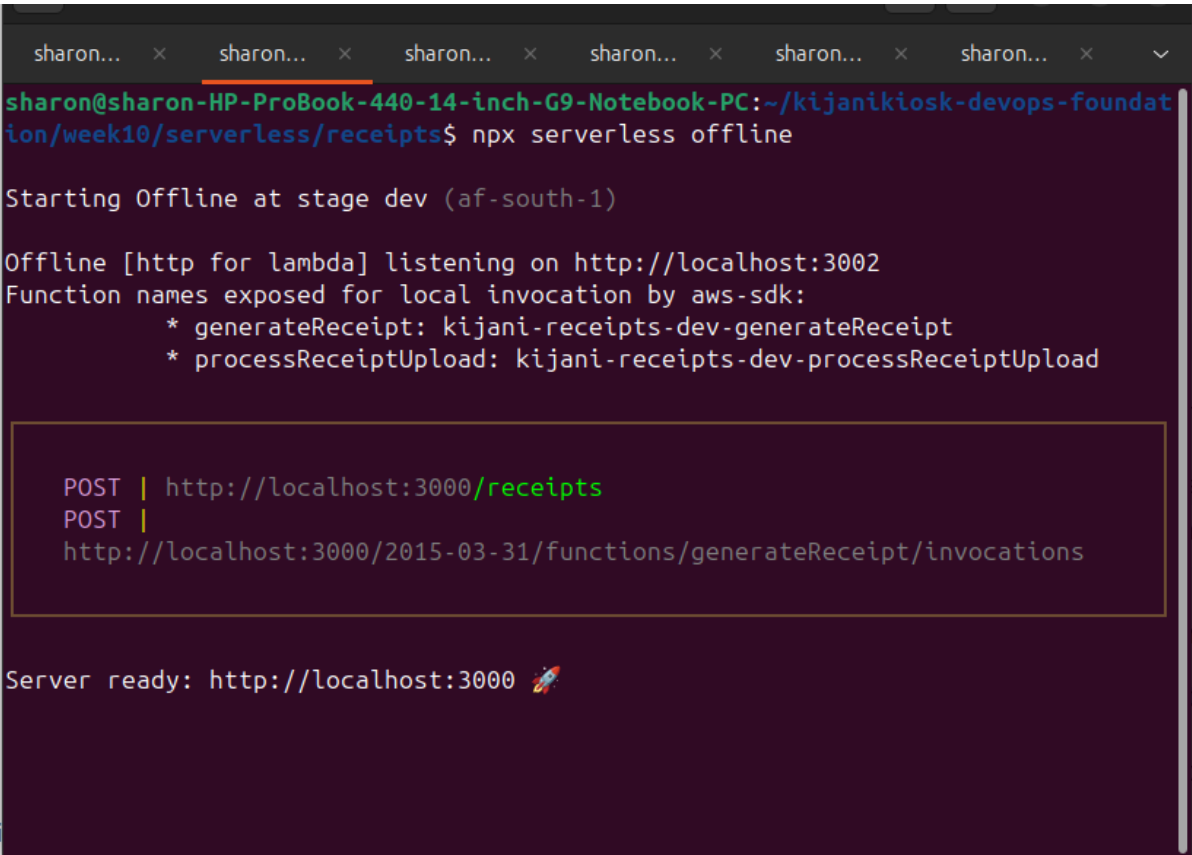
This evidence confirms recovery of the kk-payments workload, with the Kubernetes pods returning to a healthy Running and Ready state after the faulty condition was corrected.

```
ion/week10/serverless/receipts$ cd ../../..
sharon@sharon-HP-ProBook-440-14-inch-G9-Notebook-PC:~/kijanikiosk-devops-foundat
ion$ kubectl get pods -n kijani-staging
NAME                                READY   STATUS    RESTARTS   AGE
kk-api-6d874fcd7d-kc7lv             1/1     Running   19 (100m ago)   8d
kk-api-6d874fcd7d-qq6mz             1/1     Running   17 (100m ago)   8d
kk-payments-649fc85bb-dc5r6         1/1     Running   0              3m28s
kk-payments-649fc85bb-mh6bs         1/1     Running   0              3m51s
kk-payments-649fc85bb-xjl7w         1/1     Running   0              3m50s
sharon@sharon-HP-ProBook-440-14-inch-G9-Notebook-PC:~/kijanikiosk-devops-foundat
ion$
```

```
ion/week10/serverless/receipts$ cd ../../..
sharon@sharon-HP-ProBook-440-14-inch-G9-Notebook-PC:~/kijanikiosk-devops-foundat
ion$ kubectl get pods -n kijani-staging
NAME                                READY   STATUS    RESTARTS   AGE
kk-api-6d874fcd7d-kc7lv             1/1     Running   19 (100m ago)   8d
kk-api-6d874fcd7d-qq6mz             1/1     Running   17 (100m ago)   8d
kk-payments-649fc85bb-dc5r6         1/1     Running   0              3m28s
kk-payments-649fc85bb-mh6bs         1/1     Running   0              3m51s
kk-payments-649fc85bb-xjl7w         1/1     Running   0              3m50s
sharon@sharon-HP-ProBook-440-14-inch-G9-Notebook-PC:~/kijanikiosk-devops-foundat
ion$ kubectl get deployment kk-payments -n kijani-staging -o wide
NAME                                READY   UP-TO-DATE   AVAILABLE   AGE   CONTAINERS   IMAGES
kk-payments 3/3      3              3              8d    kk-payments  sharonshaz/kk
-payments:1.0.0-7234083 app=kk-payments
sharon@sharon-HP-ProBook-440-14-inch-G9-Notebook-PC:~/kijanikiosk-devops-foundat
ion$ kubectl get deployment kk-payments -n kijani-staging -o jsonpath='{.spec.tem
plate.spec.containers[0].readinessProbe.httpGet.path}'
/health
sharon@sharon-HP-ProBook-440-14-inch-G9-Notebook-PC:~/kijanikiosk-devops-foundat
ion$ kubectl rollout status deployment/kk-payments -n kijani-staging
deployment "kk-payments" successfully rolled out
sharon@sharon-HP-ProBook-440-14-inch-G9-Notebook-PC:~/kijanikiosk-devops-foundat
ion$
```

Serverless Receipt Generation

This evidence demonstrates successful receipt generation through the Week 10 serverless receipt workflow, returning HTTP 200 with the generated receipt details, including the receipt ID, order ID, amount, currency, timestamp, and generated status.



```
sharon@sharon-HP-ProBook-440-14-inch-G9-Notebook-PC:~/kijanikiosk-devops-foundation/week10/serverless/receipts$ npx serverless offline

Starting Offline at stage dev (af-south-1)

Offline [http for lambda] listening on http://localhost:3002
Function names exposed for local invocation by aws-sdk:
  * generateReceipt: kijani-receipts-dev-generateReceipt
  * processReceiptUpload: kijani-receipts-dev-processReceiptUpload

POST | http://localhost:3000/receipts
POST | http://localhost:3000/2015-03-31/functions/generateReceipt/invocations

Server ready: http://localhost:3000 🚀
```

```
sharon@sharon-HP-ProBook-440-14-inch-G9-Notebook-PC:~$ cd ~/kijanikiosk-devops-foundation/week10/serverless/receipts
sharon@sharon-HP-ProBook-440-14-inch-G9-Notebook-PC:~/kijanikiosk-devops-foundation/week10/serverless/receipts$ curl -i -X POST http://localhost:3000/receipts \
  -H "Content-Type: application/json" \
  -d '{"orderId":"TEST-001","amount":500,"currency":"KES"}'
HTTP/1.1 200 OK
content-type: application/json; charset=utf-8
cache-control: no-cache
content-length: 163
Date: Mon, 24 Aug 2026 01:23:26 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"receiptId":"fcf73e57-afd8-40d4-9588-6ea2b2c9594c","orderId":"TEST-001","amount":500,"currency":"KES","timestamp":"2026-08-24T01:23:26.693Z","status":"generated"}
sharon@sharon-HP-ProBook-440-14-inch-G9-Notebook-PC:~/kijanikiosk-devops-foundation/week10/serverless/receipts$
```

Prometheus Monitoring Stack

This evidence shows the Prometheus monitoring stack running in the Kubernetes cluster, providing the monitoring infrastructure for the application.


```
sharon@sharon-HP-ProBook-440-14-inch-G9-Notebook-PC:~/kijanikiosk-devops-foundation/week10/serverless/receipts$ kubectl get pods -A | grep -i prometheus
monitoring      alertmanager-monitoring-kube-prometheus-alertmanager-0    2/2
  Running      21 (125m ago)    8d
monitoring      monitoring-kube-prometheus-operator-54754cf8cc-h878p        1/1
  Running      36 (103m ago)    8d
monitoring      monitoring-prometheus-node-exporter-8ms9f                   1/1
  Running      19 (125m ago)    8d
monitoring      prometheus-monitoring-kube-prometheus-prometheus-0        2/2
  Running      24 (125m ago)    8d
sharon@sharon-HP-ProBook-440-14-inch-G9-Notebook-PC:~/kijanikiosk-devops-foundation/week10/serverless/receipts$
```

Prometheus Alert Rule

This evidence shows the custom KKPaymentsPodRestart alert rule deployed in Prometheus to monitor payment-service container restarts.

```

monitoring/prometheus-monitoring-kube-prometheus-prometheus-0 2/2 Running 24 (123M ago) 80
sharon@sharon-HP-ProBook-440-14-inch-G9-Notebook-PC:~/kijanikiosk-devops-foundation/week10/serverless/receipts$ kubectl get prometheusrule kk-payments-alerts -n monitoring -o yaml
apiVersion: monitoring.coreos.com/v1
kind: PrometheusRule
metadata:
  annotations:
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"monitoring.coreos.com/v1","kind":"PrometheusRule","metadata":{"annotations":{},"labels":{"release":"monitoring"},"name":"kk-payments-alerts","namespace":"monitoring"},"spec":{"groups":[{"name":"kk-payments","rules":[{"alert":"KKPaymentsPodRestart","annotations":{"description":"A kk-payments container in kijani-staging has restarted within the last 5 minutes.","summary":"kk-payments pod restarted"},"expr":"increase(kube_pod_container_status_restarts_total{namespace='kijani-staging', container='kk-payments'})[5m]) \u003e 0","for":"1m","labels":{"severity":"warning"}}]}]}
    prometheus-operator-validated: "true"
  creationTimestamp: "2026-08-15T11:28:23Z"
  generation: 1
  labels:
    release: monitoring
  name: kk-payments-alerts
  namespace: monitoring
  resourceVersion: "179629"
  uid: ef42b1c0-a3e3-4c61-afa6-999e78a4e829
spec:
  groups:
  - name: kk-payments
    rules:
    - alert: KKPaymentsPodRestart
      annotations:
        description: A kk-payments container in kijani-staging has restarted within the last 5 minutes.
        summary: kk-payments pod restarted
      expr: increase(kube_pod_container_status_restarts_total{namespace="kijani-staging", container="kk-payments"})[5m] > 0
      for: 1m
      labels:
sharon@sharon-HP-ProBook-440-14-inch-G9-Notebook-PC:~/kijanikiosk-devops-foundation/week10/serverless/receipts$

```

Prometheus Alert Evaluation

This evidence confirms that Prometheus has successfully loaded and evaluated the KKPaymentsPodRestart rule. The rule is currently inactive because no qualifying payment pod restart is occurring.

```
sharon@sharon-HP-ProBook-440-14-inch-G9-Notebook-PC:~/kijanikiosk-devops-foundat
ion$ curl -s http://localhost:9090/api/v1/rules | python3 -c '
import sys,json
d=json.load(sys.stdin)
for g in d["data"]["groups"]:
    for r in g["rules"]:
        if r["name"]=="KKPaymentsPodRestart":
            print(json.dumps({
                "state": r["state"],
                "name": r["name"],
                "health": r["health"],
                "lastEvaluation": r["lastEvaluation"],
                "alerts": r["alerts"]
            }, indent=2))
'
{
  "state": "inactive",
  "name": "KKPaymentsPodRestart",
  "health": "ok",
  "lastEvaluation": "2026-08-24T02:40:13.177477001Z",
  "alerts": []
}
sharon@sharon-HP-ProBook-440-14-inch-G9-Notebook-PC:~/kijanikiosk-devops-foundat
ion$
```

ori

+